



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
PRE BOARD EXAMINATION

CLASS: X
SUBJECT: CHEMISTRY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

Maximum Marks: 80 Time allowed: Two hours

Answers to this Paper must be written on the paper provided separately. You will not be allowed to write during first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Section A is compulsory. Attempt any four questions from Section B. The intended marks for questions or parts of questions are given in brackets [].
(This paper consists of seven printed pages)

SECTION A

(Attempt all questions from this Section)

Question 1

Choose the correct answers to the questions from the given options.

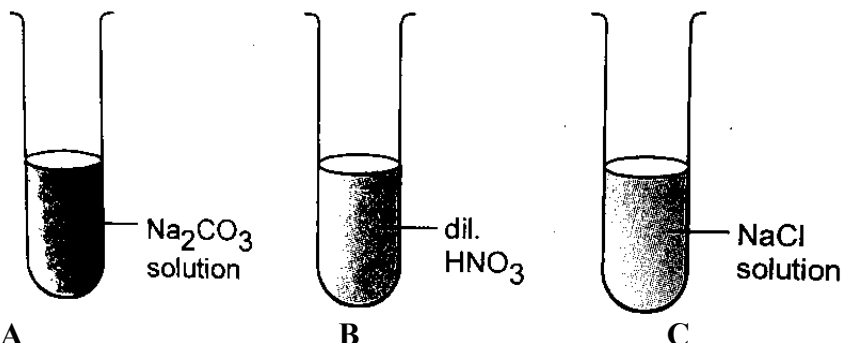
[15]

(Do not copy the questions, write the correct answers only.)

- (i) A metal spoon is to electroplated with nickel. Which of the following could be the electrodes?
P. anode: spoon; cathode: nickel
Q. anode: nickel; cathode: nickel
R. anode: nickel; cathode: spoon
(a) both P and Q
(b) only Q
(c) only P
(d) only R
- (ii) A student did an activity in which he added sodium bicarbonate to hydrochloric acid. The gas released was passed through lime water. What change would be observed in lime water?
(a) bubbles were formed
(b) white turbid was formed
(c) the solution became colourless
(d) the colour of the solution became red
- (iii) Which of the following has higher number of molecules?
(a) 20 gram of oxygen
(b) 5 gram of nitrous oxide
(c) 2.24 litres of sulphur dioxide at STP
(d) 2 gram atoms of chlorine [At. Wt:- O=16, N=14, S=32, Cl=35.5]
- (iv) The reaction taking place at anode during silver plating is:
(a) $4\text{OH}^- - 4\text{e}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
(b) $\text{Ag} - \text{e}^- \rightarrow \text{Ag}^+$
(c) $2\text{CN}^- - 2\text{e}^- \rightarrow 2\text{C} + \text{N}_2$
(d) $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$

- (v) Assertion (A): Collection of hydrogen chloride gas in the laboratory preparation method is collected by the downward delivery.
Reason (R): Hydrogen chloride is 1.28 times heavier than air.
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true
- (vi) Which of the following is an isomer of 2-methyl butane :
(a) n- butane
(b) propane
(c) n- pentane
(d) but-2-ene
- (vii) A polar covalent bond will be formed in which one of these pair of atoms is:
(a) hydrogen
(b) methane
(c) hydrogen fluoride
(d) oxygen
- (viii) An element 'P' has the value of Z as 16. The period and group to which P belongs respectively is:
(a) 3, 16
(b) 1, 2
(c) 6, 16
(d) 3, 6
- (ix) When Na_2CO_3 solution is added to CuSO_4 solution, the colour of the precipitate is:
(a) blue
(b) light green
(c) brown
(d) white
- (x) The element having highest ionization energy is:
(a) lithium
(b) carbon
(c) fluorine
(d) helium
- (xi) Assertion (A): 'P' is a weak acid having low concentration of hydronium ions.
Reason (R) : 'P' contains ions.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
- (xii) The ratio between the number of molecules in 2 g hydrogen and 32 g oxygen is:
[O = 16]
(a) 0.01:1
(b) 1: 1
(c) 1:0.01
(d) 1:2
- (xiii) When a salt X is heated with copper and conc. sulphuric acid, it liberates a reddish brown gas. The anion present in 'X' is:
(a) carbonate
(b) nitrate
(c) sulphate
(d) sulphite

- (xiv) The incorrect statement about a coordinate bond is:
- It is formed by a shared pair of electrons coming from one atom
 - It behaves as a covalent bond
 - Number of coordinate bond is equal to the number of lone pair of electrons
 - It is caused by bond pair effect
- (xv) Which of the following is correct about the colour of pH paper and the nature of the solution in test tubes A, B and C respectively.

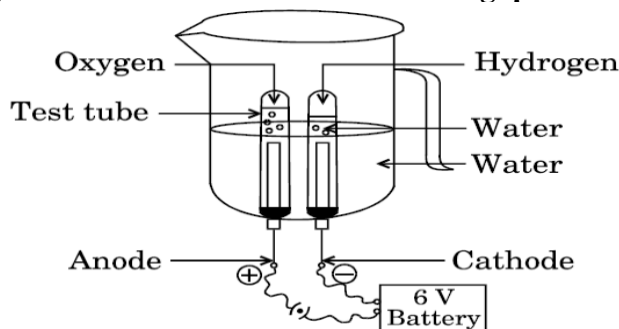


- A- red, basic; B- red, acidic; C- yellowish green, neutral
- A- blue, basic; B- red, acidic; C- yellowish green, neutral
- A- red, acidic; B- blue, basic; C- blue, neutral
- A- blue, acidic; B- red, basic; C- green, neutral

Question 2

- (i) Study the diagram given below and answer the following questions:

[5]



- Name the process depicted in the diagram.
- State the composition of the electrodes
- The reaction does not take place if a few drops of dilute sulphuric acid are not added to water. Why?
- Write balanced chemical reactions taking place at the respective electrodes.

- (ii) Match the following column A with column B:

[5]

Column A	Column B
(a) Non electrolyte	1. Ammonium chloride
(b) Three types of bond	2. Distilled water
(c) Carbonate ore	3. Carbon to carbon dioxide
(d) Concentrated sulphuric acid	4. High solubility of a gas
(e) Fountain experiment	5. Calcination

- (iii) Complete the following by choosing the correct answers from the bracket:

[5]

- If a metallic oxide is dissolved in water, the resulting solution has a higher concentration of ____ (OH⁻/ H⁺)
- A triacidic base is ____ (phosphoric acid/ aluminium hydroxide).
- Ferric chloride cannot be prepared by ____ (simple displacement/ direct combination).

(d) A molecule having triple covalent bond is ____ (nitrogen/ ammonia).

(e) A white precipitate soluble in excess ammonium hydroxide is ____ (PbCl_2 / AgCl / BaSO_4)

(iv) Write the suitable term for the following: [5]

(a) The method used to concentrate zinc blende.

(b) The number of particles present in one mole of a substance.

(c) Process of heating a concentrated ore in excess supply of air.

(d) The number of replaceable hydrogen ions produced per molecule of the acid in aqueous solution.

(e) Type of reactions ethane undergoes.

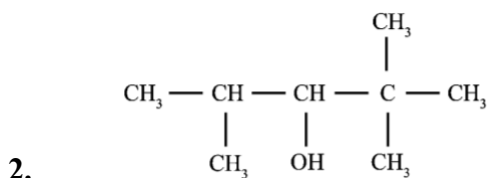
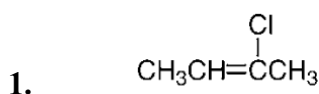
(v) (a) Draw the structural diagram for the following compounds: [5]

1. The first homologue of ketone series

2. A carboxylic acid having five carbon atoms

3. Positional isomer of butanol

(b) Give the IUPAC name of the following organic compounds:



SECTION B (40 Marks)
(Attempt any four questions)

Question 3

(i) Identify the anion present in the compound 'X' and write the balanced equation for the following: [2]

Sulphuric acid when warmed with a compound 'X' gives off a colourless gas which turns acidified potassium dichromate solution green.

(ii) What property of Nitric acid(dilute or concentrated) is exhibited in each of the following cases: [2]

(a) In the preparation of carbon dioxide from carbon.

(b) In the preparation of nitric oxide from copper.

(iii) The electronegativity of an element X is lesser than that of element Y. [3]

(a) Which one has more atomic radius?

(b) How is the ionization potential of X likely to compare with that of Y?

(c) State whether X is likely to be placed to the right or to the left of Y in the periodic table.

(iv) You are provided with the list of chemicals mentioned below in the box: [3]

Nitric acid, caustic soda solution, Zinc sulphate, zinc, nitrogen dioxide, lead, hydrochloric acid, ammonium carbonate, lead hydroxide, sulphuric acid, carbonic acid

Using suitable chemicals from the list given, write balanced chemical equation for the preparation of salts mentioned below: (Using one reagent only once) [3]

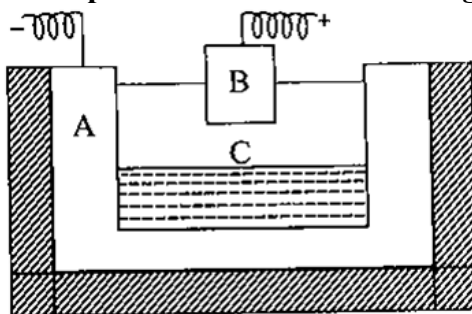
(a) Zinc carbonate

(b) Sodium plumbite

(c) Lead nitrate

Question 4

- (i) The following questions relate to the industrial preparation of nitric acid: [2]
 (a) Name the process.
 (b) Give balanced chemical equation for the conversion of nitrogen dioxide to nitric acid.
- (ii) On analyzing an organic sample weighing 0.5 grams, Ravi found the sample contains 0.062 grams of hydrogen and 0.25 gram of oxygen (atomic weight 16). Determine the empirical formula of the compound obtained by Ravi. [2]
- (iii) Give balanced equation for each of the following: [3]
 (a) Passage of ammonia and oxygen over heated platinum.
 (b) Oxidation of sulphur using sulphuric acid.
 (c) Addition of dilute nitric acid to zinc granules.
- (iv) The diagram below is a representation of an electrolytic cell used in the extraction of aluminium. With respect to the process answer the following questions: [3]



- (a) Identify the element of which the electrodes A and B are made .
 (b) At which electrode (A or B) is aluminium formed?
 (c) C, an electrolyte contains two aluminium compounds. One is alumina, name the other.

Question 5

- (i) (a) Anjali wants to prove that ammonia is a good reducing agent. To demonstrate this, she passed excess chlorine through ammonia. What did she observe?
 (b) Write a balanced chemical equation for the above reaction. [2]
- (ii) Name the alloy which is made up of: [2]
 (a) aluminium, magnesium, manganese, copper
 (b) Lead, tin
- (iii) Rohit was given a salt 'X' which was white in colour for analysis. On strong heating it produced a residue which is yellow when hot and white when cold, a colourless gas and a reddish brown gas. The solution of salt 'X' when tested with excess of ammonium hydroxide produced a colourless solution.
 (a) Name the salt taken by Rohit.
 (b) Write balanced chemical equations to show the reactions when 'X' is treated with excess ammonium hydroxide. [3]
- (iv) Given below in column A is a schematic diagram of the concentration of ore. Identify the parts labelled as A, B and C with the correct options from the column B. [3]

Column A		Column B
	1.	Magnetic wheel
	2.	Non magnetic wheel
	3.	Magnetic ore
	4.	Gangue
	5.	Pulverised ore

Question 6

- (i) Element 'X' forms a nitride with formula 'XN', which is a solid with high melting point. 'X' would most likely be placed in the group of the Periodic Table as:

1. (a) Li (b) Ca (c) B (d) C

2. Justify the position of 'X' in the periodic table.

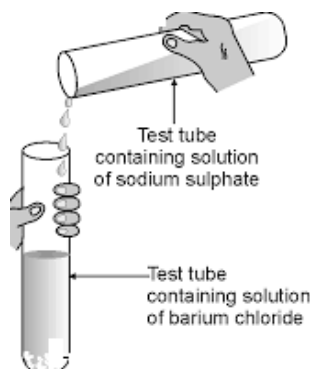
[2]

- (ii) A student was asked to perform two experiments in the laboratory based on the Instructions given:

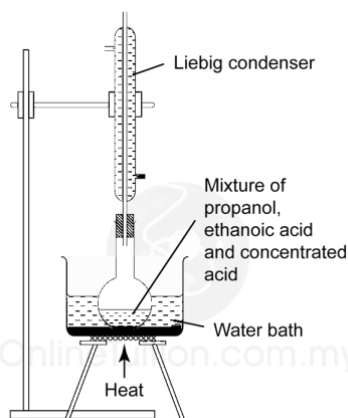
Observe the picture given below and state one observation for each of the Experiments 1 and 2 that you would notice on mixing the given solution.

[2]

(a) Experiment 1



(b) Experiment 2



- (iii) Copper is electrorefined using copper sulphate solution.

[3]

(a) Which electrode has the impure block of copper?

(b) Write the equation for the reaction occurring at the above electrode.

- (iv) X [2, 8, 8, 7] and Y [2, 8, 2]. Using this information complete the following:

[3]

(a) _____ is the non-metallic element.

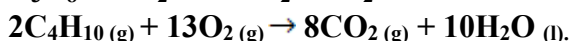
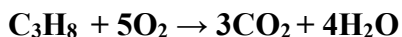
(b) The formula of the compound formed between X and Y is _____.

(c) _____ is the oxidising agent.

Question 7

- (i) One variety of household fuel is a mixture of propane (30%) and butane (70%). If 40 litres of this mixture is burnt, find the total volume of carbon dioxide added to the atmosphere.

The combination reactions can be represented as:



State the law used for the above calculations.

[3]

- (ii) Karan has solutions X, Y and Z that has pH 7, 13 and 3 respectively. Which solution:

[3]

(a) will liberate a pungent smelling gas when reacted with ammonium carbonate.

(b) will turn anhydrous copper sulphate blue.

(c) will produce a colourless gas that turns lime water milky but no effect on acidified

potassium permanganate solution?

- (iii) 0.02 moles of pure manganese oxide is heated strongly with conc hydrochloric acid according to the equation



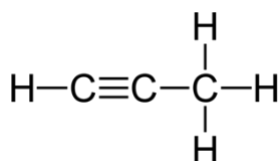
Calculate the following:

[4]

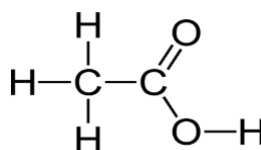
- (a) Moles of the salt formed.
- (b) Volume of chlorine gas formed at STP.
- (c) mass of the acid required.
- (d) mass of the salt produced. [Mn=55, Cl=35.5, O=16]

Question 8

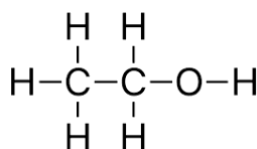
- (i) State giving reasons if: [2]
 - (a) Calcium nitrate and magnesium nitrate can be distinguished using sodium hydroxide solution.
 - (b) Sodium sulphate and ammonium sulphate solids can be distinguished by a solution having pH 12.
- (ii) Draw the electron dot diagram of ammonium ion. [2]
- (iii) Give balanced equations for the following: [3]
 - (a) Laboratory preparation of ethyne by dehydrohalogenation.
 - (b) Conversion of methyl iodide to methane
 - (c) Laboratory preparation of hydrogen chloride.
- (iv) The structures of five organic compounds are shown: [3]



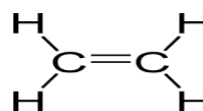
A



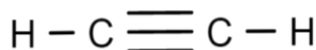
B



C



D



E

- (a) Identify two of the compounds that are members of the same homologous series but not isomers to each other.
- (b) Which compound in its aqueous state turns blue litmus red?
- (c) 'D' can be prepared from 'C'. Give a chemical equation for the reaction.